



How We Tested

Compatibility among applications can be judged on two grounds: functional and procedural. Functional compatibility is the ability of the software to emulate the functional capabilities of other software. Procedural compatibility is the software's ability to enable one to do the same job, using similar steps. Functional compatibility is required to ensure that the software works with the existing application. Procedural compatibility, on the other hand, reduces the time required to re-train oneself when migrating to the new application. Here, we only tested the functional compatibility of the alternative suites with MS Office. Hence, the testing concentrated on the ability of the alternative suites to open existing Word, Excel and PowerPoint files, make changes to them, save them and finally, let them open again in MS Office. The files were created to take into consideration the limitations that vendors face while ensuring MS Office compatibility. We therefore included only a bare minimum of the set of features that MS Office offers, compatibility with which is essential in any office environment. The more advanced of features were left out.

The test documents came under two categories: ones that were created by us, and ones that were downloaded from the Internet. The ones downloaded from the Internet contained a lot of graphical diagrams and mathematical equations. The mathematical equations range from very simple ones to highly complex nested equations.

Each document that opened without any errors scored five points; those with minor errors scored three points; those with major errors, but which were still operable got one point; while documents which were unopenable scored zero for the suite.

Word Processors: In the case of word processors, text formatting is very important. So, the main test document consisted of text written in a variety of fonts, with varying font sizes and formatting. In certain places, the background and foreground colours were reversed. Almost all the other test documents had images, screenshots and diagrams, and the ability of the word processor to display these was tested. In total we tested each suite with 10 documents. Some screenshots had callouts describing the various components, and

the ability of the applications to render these was also tested. In total, there we're

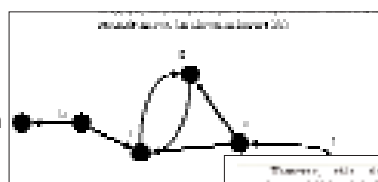
Spreadsheets: Here, correct reproduction of formulas and embedded graphs was given top priority. We used nine documents to test this category. Many of the test documents were designed to test the ability of the applications to open multiple worksheets in a single spreadsheet. These had formulas referencing multiple worksheets, and also used graphs that were cross-referenced to other worksheets. One test spreadsheet had complex functions and basic Windows widgets (simple controls such as buttons) embedded. We didn't expect it to work on Linux-based applications—they were specifically meant for Windows.

Presentations: The PowerPoint presentation files were used to check the ability of the alternative presentation tools to correctly display the content of the slides. Both the quality, as well as the precision of the displayed slides, was evaluated. We used 12 presentations to test this category, one of which was designed to check the ability of the software to reproduce colours and transition effects. Another presentation tested graph capabilities. This presentation consisted of various types of graphs. It starts off

Classification of errors: Errors in documents were classified as fine, minor, major and unusable. Documents displayed with hardly any errors were listed as 'fine'; those with very few mistakes were listed as 'minor'. Slightly more complex errors that spoil the user experience were classified as 'major'. 'Unusable' errors were ones that rendered the document unusable.

For example, in the case of presentations, if the background image was not displayed in the same way as in the original, the error was considered as 'major', since the purpose

In word processors, slightly misplaced images were given less importance than completely misplaced ones



of putting a background image in a presentation is to reduce its prominence, and to give a subtle effect.

In the case of Word documents, the rules were slightly different. If, for example, an image was slightly misplaced, the error was classified as 'minor'. But if it was completely misplaced and, say, overlapped with the



Background images were given more importance while evaluating presentation software

Item	Item Name	Item Value
1	Item 1	100.00
2	Item 2	200.00
3	Item 3	300.00
4	Item 4	400.00
5	Item 5	500.00
6	Item 6	600.00
7	Item 7	700.00
8	Item 8	800.00
9	Item 9	900.00
10	Item 10	1000.00

Merged cells were given more importance since they indicate major categories, and contribute to the overall impact of the data

text, it was considered 'unusable'. However, in documents where the images were simple but highly relevant to the text, the absence of even a single image was considered an 'unusable' error.

In the case of spreadsheets, the ability of the software to correctly understand formulae and display embedded graphs was given more importance than minor formatting errors. But certain places where formatting was more relevant, such as cells merged to indicate common data under a single heading, were given high importance.

with simple bar graphs, and proceeds towards more complex graphs. In each graph type, both a 2D as well as a 3D version of the same graph was used. Several slides had embedded foreground and background images, along with text displayed using sliding effects, and the correct reproduction of these was given weightage.